

Section 6.1a – Equation of a Line in Slope y-intercept form

MPM1D

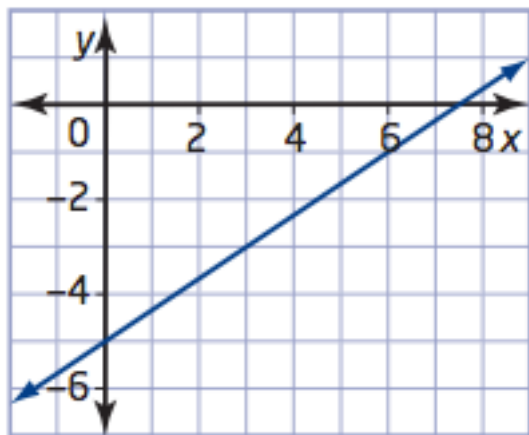
Learning Goals: In this lesson students will be able to determine the equation of a line in the form $y = mx + b$ from its graph. They will be able to use the formula $\frac{y_2 - y_1}{x_2 - x_1}$ to calculate the slope of a line.

Example 1: Complete the following chart

Equation	Slope	y-intercept
$y = -2x - 5$		
$y = x + 2$		
$y = \frac{2}{5}x + 8$		
$y = -\frac{1}{2}x$		
$y = 4$		

Example 2: Identify the slope and y-intercept of each line

a)



Slope: to find the slope use two points on the line and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

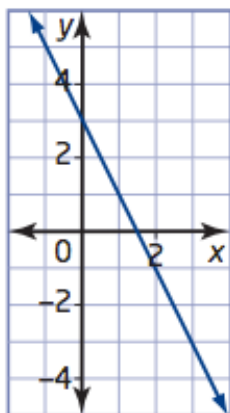
y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis. (When $x = 0$, $y = ?$)

Slope:

y-intercept:

Equation of the line:

b)



Slope: to find the slope use two points on the line

and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

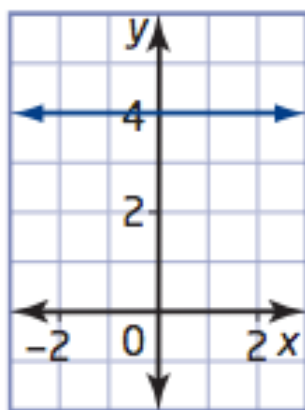
y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis. (When $x = 0$, $y = ?$)

Slope:

y-intercept:

Equation of the line:

c)



Slope: to find the slope use two points on the line

and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis. (When $x = 0$, $y = ?$)

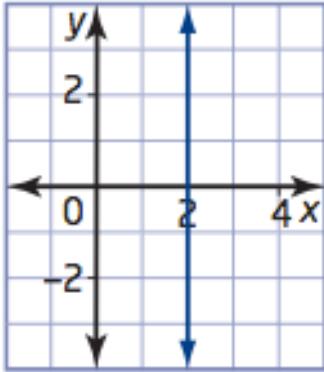
Slope:

y-intercept:

Equation of the line:

Note: all horizontal lines have a slope of _____ and an equation of the form _____, where h is the

d)



Slope: to find the slope use two points on the line
and the formula $m = \frac{\text{rise}}{\text{run}}$ OR $m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$

y-intercept: you can find the y-intercept by looking at the graph and checking where the line crosses the y-axis.

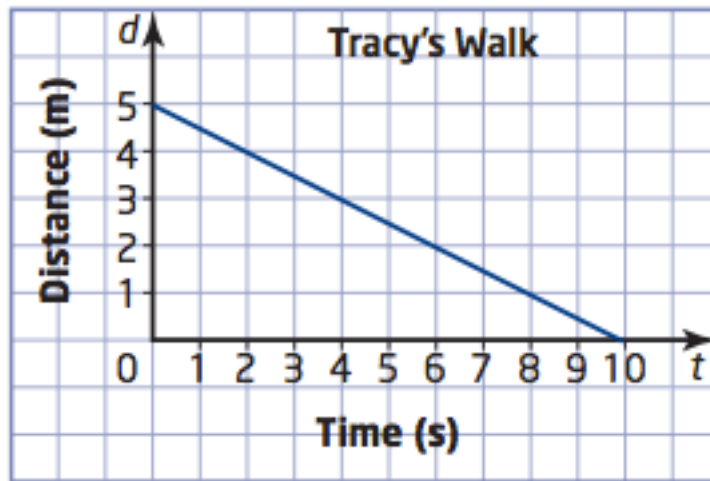
Slope:

y-intercept:

Equation of the line:

Note: All vertical lines have an _____ slope and an equation of the form _____, where a is the _____.

Example 3: Interpreting a Linear Relation



Identify the slope and the vertical intercept of the linear relation and explain what they mean.

Slope:

y-intercept:

The slope represents Tracy's _____. The negative value means that her distance from the sensor is _____. Tracy's speed toward the sensor was _____.

y-intercept:

The y-intercept of _____ means that Tracy started walking at a distance of _____ from the sensor.

Equation of the relation: